SIADS 699 Capstone Project Report

SafePath

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Introduction

Traffic crashes in New York City occur within a complex system shaped by the physical structure of streets, daily activity patterns, and changing environmental conditions. Understanding this system requires more than a map of past incidents. It requires tools that can identify where risk is concentrated, how different types of street behave, and when citywide conditions shift in ways that influence crash likelihood.

Cities already publish dashboards that summarize crash information. Examples include the Ann Arbor Traffic Crash Dashboard and NYC Vision Zero reporting tools. These platforms help users explore patterns by location, time, and contributing factors, and they support broader safety goals. Their focus, however, is descriptive. They do not incorporate predictive modeling, structural analysis, or temporal state detection.

Research has explored these perspectives individually. Supervised learning models have been used to identify high-risk locations (Ren et al., 2018; Nikolaou et al., 2023; Xiao et al., 2024). Clustering methods have revealed meaningful roadway segment types (Abdalazeem and Oke, 2025; Khalfa and Hardyns, 2025). Weather-driven crash risk has been documented at broader scales (Perrels et al., 2015; USDOT FHWA, 2024). Contextual features have also been shown to improve predictive performance (Hakyemez and Badur, 2026). Although these approaches provide valuable insights, they typically operate in isolation. Practitioners rarely have tools that connect predictive, structural, and temporal perspectives in a single, interpretable system.

SafePath is designed as an exploratory tool that brings these ideas together. It integrates a supervised LightGBM model that ranks street segments by predicted crash risk, a Gaussian Mixture Model that identifies structural segment archetypes, and a Hidden Markov Model that detects daily temporal regimes driven by weather and recent crash activity. The goal is not to provide definitive predictions or recommendations. Instead, SafePath offers a way to examine how spatial structure, temporal conditions, and historical patterns relate to crash risk, and to present these insights in a dashboard that supports exploration and prioritization.

Data and Feature Engineering

SafePath uses three main datasets: NYC crash records, the LION street network, and daily weather observations from Open-Meteo. Each dataset was cleaned and processed into a unified segment-day panel. A full list of features can be found in the variable dictionary in **Appendix D**.

NYC Crash Data

Crash records were downloaded from the Motor Vehicle Collision dataset on NYC Open Data, covering 2018 through 2024. Records missing geographic coordinates or valid timestamps were removed, and the date and time fields were combined into a single datetime column.

Each crash was assigned to a street segment using a spatial join with the LION network. Crashes falling outside a small buffer around any segment were dropped. After cleaning and spatial assignment, approximately 958,000 crashes were linked to a 'segment_id'. Daily crash counts and a binary crash indicator were computed for each segment.

LION Roadway Data

The LION dataset provides the geometric and structural attributes of New York City's street network. Segment-level variables include road class, number of lanes, speed limits, one-way direction, and bridge or tunnel indicators. Node attributes were used to identify intersections, traffic signals, and pedestrian crossings.

Weather Data

Daily weather observations were obtained from the Open-Meteo archive for the 2018–2024 period. Variables include daily mean temperature(C) measured at 2 meters, total precipitation(mm), and maximum wind speed(km/h) measured at 10 meters. Weather data was merged into the segment-day panel by date.

Feature Engineering

Feature engineering was performed at two levels: segment-day features and segment-level aggregates.

Segment-Day Features

The segment-day panel contains one row per segment per day. For each row, we computed the daily crash count and a binary crash_occurred flag. Rolling 7-day and 30-day crash totals were added to capture recent activity, and these windows were shifted to avoid leakage from the current day.

Weather variables from Open-Meteo were merged by date, and a rain indicator was created from daily precipitation totals. Calendar fields such as day of week and month were added, along with cyclical encodings to represent seasonal patterns.

Infrastructure features derived from the LION dataset were merged into the panel. These include curvature, maximum change in bearing, and intersection degree, which together capture geometric complexity and local intersection density. A visibility-related score was constructed as a weighted combination of normalized curvature (0.4), bearing change (0.4), and intersection complexity (0.2). The heavier weights on curvature and bearing reflect their stronger impact on how far ahead a driver can see, while intersection complexity plays a smaller, secondary role.

Additional attributes such as lane count, speed limit, and structural flags (bridge, tunnel, one-way street) were also included. To capture broader exposure patterns, we also computed citywide rolling averages of crash counts over 7-day and 30-day windows. These aggregate features provide temporal context for segment-level activity and support the downstream modeling of day-to-day fluctuations in crash risk.

Segment-Level Features

For the clustering analysis, crash behavior was aggregated across the full study period for each segment. These aggregates include total crashes, average crash rate, crash volatility, percentage of days with a crash, and average values of the lagged crash features. Infrastructure attributes were merged into this table as well. All features were standardized prior to clustering to ensure that variables with larger scales did not dominate the model.

Modeling Approach

SafePath uses three complementary modeling components to examine crash risk from different perspectives. The supervised model ranks segment-day risk, the clustering model identifies structural segment types, and the temporal model captures day-to-day shifts in citywide conditions. Each model contributes a different layer of insight, and together they provide a more complete view of how spatial structure, temporal patterns, and environmental conditions relate to crash risk. Full hyperparameter settings for all three models are provided in **Appendix A**.

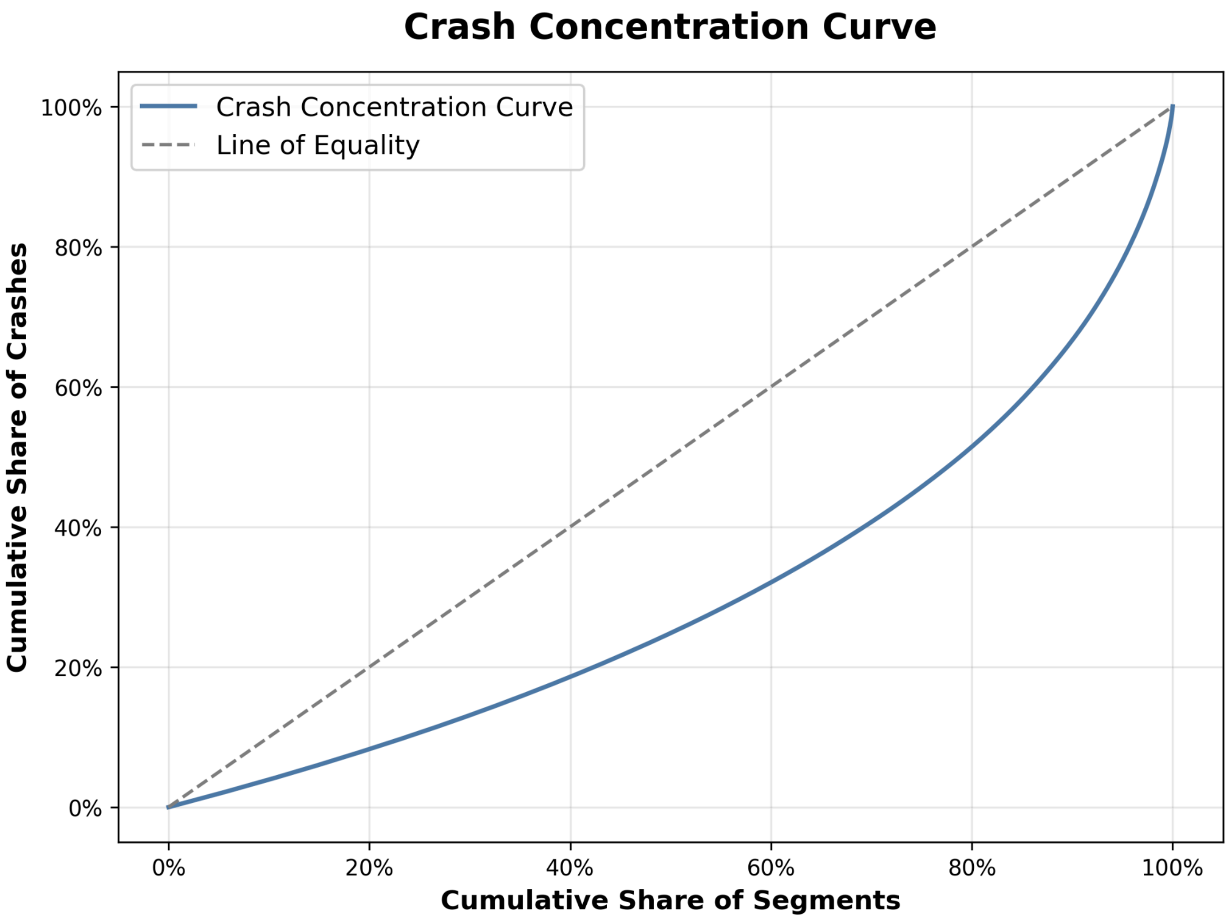
LightGBM

The supervised component uses a LightGBM classifier to estimate the relative likelihood of a crash occurring on each segment on each day. LightGBM was selected because it handles nonlinear relationships, interactions among features, and a large number of predictors without needing extensive preprocessing. The model was trained on the segment-day panel using infrastructure features, weather variables, temporal indicators, and recent crash history.

To evaluate the model on future conditions, the dataset was split chronologically. Earlier years were used for training, the following year was used for validation, and the final year was held out for testing. This approach prevents information from later periods from influencing earlier ones and reflects how the model would perform in a real-world setting.

A central consideration in designing the supervised model is that crashes are rare events. Most segments experience no crashes on most days, and even over long periods, crash activity is highly concentrated. Figure 1 shows a Lorenz-style concentration curve for the NYC street network, illustrating the cumulative share of crashes accounted for by the cumulative share of segments. If crashes were evenly distributed, the curve would follow the diagonal. Instead, it bows sharply below the line of equality, indicating that a relatively small portion of the network contributes disproportionately to total crash activity.

Figure 1. Lorenz-Style Concentration Curve for NYC Street Segments



This structural imbalance has important implications for model evaluation. Traditional classification metrics such as accuracy or precision are not informative in this setting because the overwhelming majority of segment-days contain no crash. A model could achieve high accuracy by predicting “no crash” everywhere, and precision is difficult to interpret when positive events are extremely sparse.

For these reasons, the supervised model is used primarily as a ranking tool rather than a traditional classifier. The goal is to prioritize the segments most likely to experience elevated risk on a given day, enabling planners to focus attention on a manageable subset of the network. This motivates the use of ranking-based evaluation metrics, such as Recall@K, which assess how effectively the model identifies the highest-risk segments within a specified portion of the network. Similar ranking-focus evaluations are used in other segment-level crash-risk studies (Ren et al., 2018; Nikolaou et al., 2023).

Gaussian Mixture Model

To understand long-term structural patterns, a Gaussian Mixture Model (GMM) was applied to segment-level aggregates. These aggregates include crash rate, crash volatility, percentage of days with a crash, and key infrastructure features. The GMM groups segments that share similar geometrics and behavioral characteristics.

Unlike the supervised model, the GMM does not vary day-to-day. It provides a stable set of segment archetypes that reflect underlying roadway structure. These clusters help explain why certain segments consistently exhibit higher or lower risk and provide context for interpreting the supervised model's predictions.

Hidden Markov Model

Crash risk also varies with broader citywide conditions. To capture these dynamics, a Hidden Markov Model (HMM) was trained on daily weather variables and recent crash activity aggregated across the city. The HMM identifies latent regimes that represent typical patterns of conditions, such as dry-weather days with stable crash activity or days with elevated risk driven by precipitation and recent crashes.

The model also estimates transition probabilities between regimes, which show how often the city moves between different states and how persistent each state tends to be. These regimes help explain why certain days show higher predicted risk across many segments and provide an additional perspective on how crash conditions shift over time.

Results

LightGBM Results

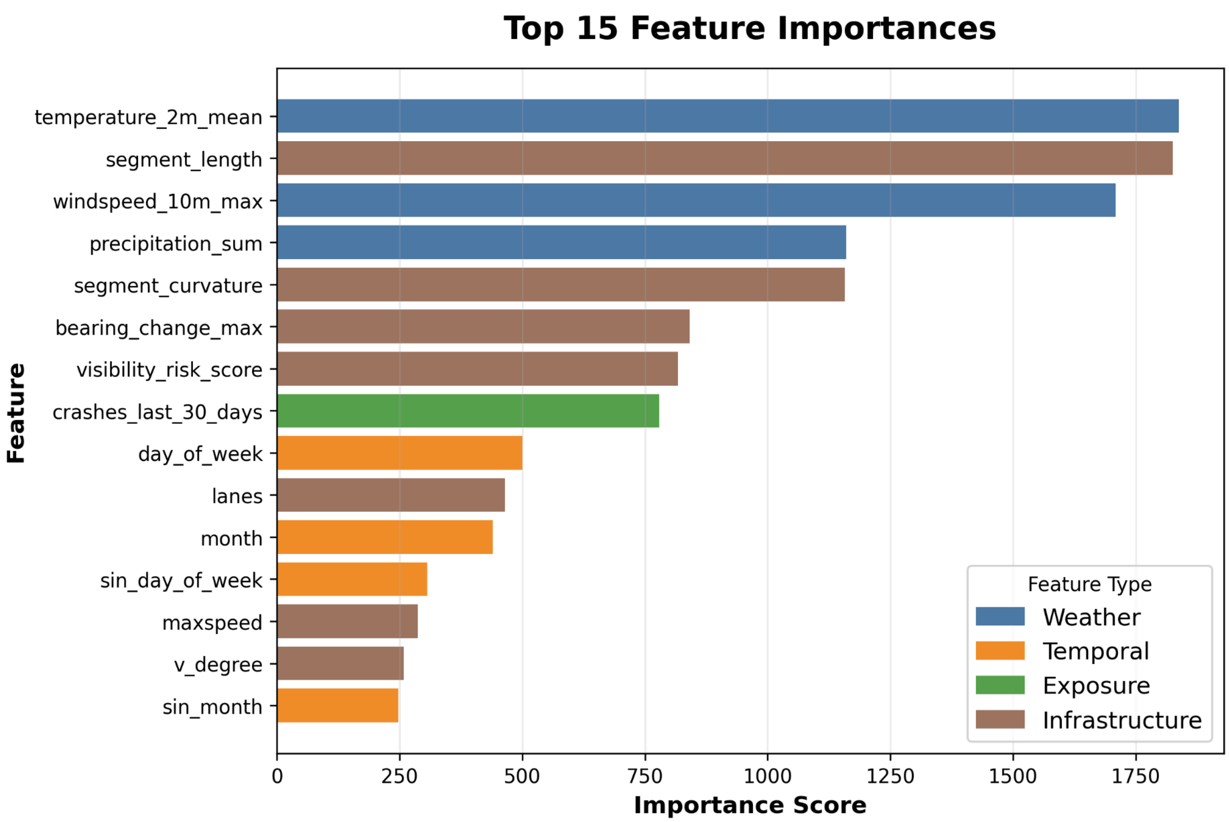
The LightGBM model provides daily, segment-level estimates of crash risk, producing a ranked list of segments for each day in the test period. Because SafePath is designed to surface the highest-risk segments for limited daily attention, we evaluate the model primarily on its ability to concentrate crashes within the top portion of predicted-risk segments. On the 2024 test set, the top 1% of predictions captures roughly 5.7% of crashes, and the top 5% captures about 17.1%, as shown in Table 1. These Recall@K values are the most operationally meaningful metrics for SafePath, as they indicate how effectively the model prioritizes segments for proactive review.

Table 1: LightGBM Top-K Recall Performance for High-Risk Segment Identification

Set	**K**	**K_rows**	**Precision @K**	**Captured Positives**	**Total Positives**	**Recall @K**
Validation	1%	42489	0.062628	2661	52460	0.050724
	5%	212448	0.040203	8541	52460	0.162810
Test	1%	42606	0.061517	2621	46239	0.056684
	5%	213030	0.037154	7915	56239	0.171176

To provide additional transparency, Figure 2 presents the top fifteen features ranked by their importance in the model. The highest scoring predictors include a mix of recent crash history, weather conditions, roadway geometry, and temporal indicators. This pattern reflects the model's use of both short-term conditions and longer-term structural characteristics when ranking segment-level risk. Similar studies have also found that combining structural and short-term features have been shown to improve predictive performance (Hakyemez and Badur, 2026).

Figure 2. Key Predictors Driving LightGBM Risk Estimates



Traditional classification metrics such as ROC-AUC, average precision, balanced accuracy, and F1 score are included in **Appendix B**. These metrics provide additional context for model performance, although they are less directly tied to SafePath's operational goal of ranking segments for daily review. The combination of Recall at K performance and feature importance patterns provides a clear picture of how the model behaves and how its predictions should be interpreted alongside the GMM and HMM components.

GMM Cluster Results

The Gaussian Mixture Model groups street segments into five long-term structural archetypes based on crash rate, geometric features, and exposure-related characteristics. These clusters do not vary day-to-day. Instead, they describe persistent differences in the built environment that shape how segments behave over time.

Two clusters exhibit consistently elevated crash rates. One cluster (Cluster 2) reflects segments with sharp curvature and limited visibility, where constrained sightlines and complex street geometry contribute to persistent risk. Another (Cluster 5), by contrast, consists of straighter segments with better visibility and fewer intersections. These segments likely represent higher-speed or higher-volume corridors where traffic density drives elevated crash risk. The remaining clusters represent typical city segments with moderate risk or low-complexity segments with the lowest crash rates. A full cluster summary table is provided in **Appendix C**.

HMM Regime Results

The Hidden Markov Model identifies three latent daily “risk regimes” based on weather conditions and recent crash activity. These regimes summarize broader environmental patterns that shape how crash risk tends to change across the city. A full summary of regime characteristics is provided in **Appendix C**.

The High-Risk Regime reflects days with elevated crash activity, characterized by higher recent crash averages, higher crash rates, and slightly stronger winds. On these days, elevated segment-level scores may indicate system-wide conditions rather than isolated hotspots. The Low-Risk Regime captures days with heavy precipitation but lower crash activity, likely reflecting reduced traffic volumes during sustained rainfall. The Moderate-Risk Regime represents typical dry-weather days with normal traffic patterns and serves as the city’s baseline operating state. Although the Low- and Moderate-Risk regimes have similar crash metrics, they arise from different underlying conditions.

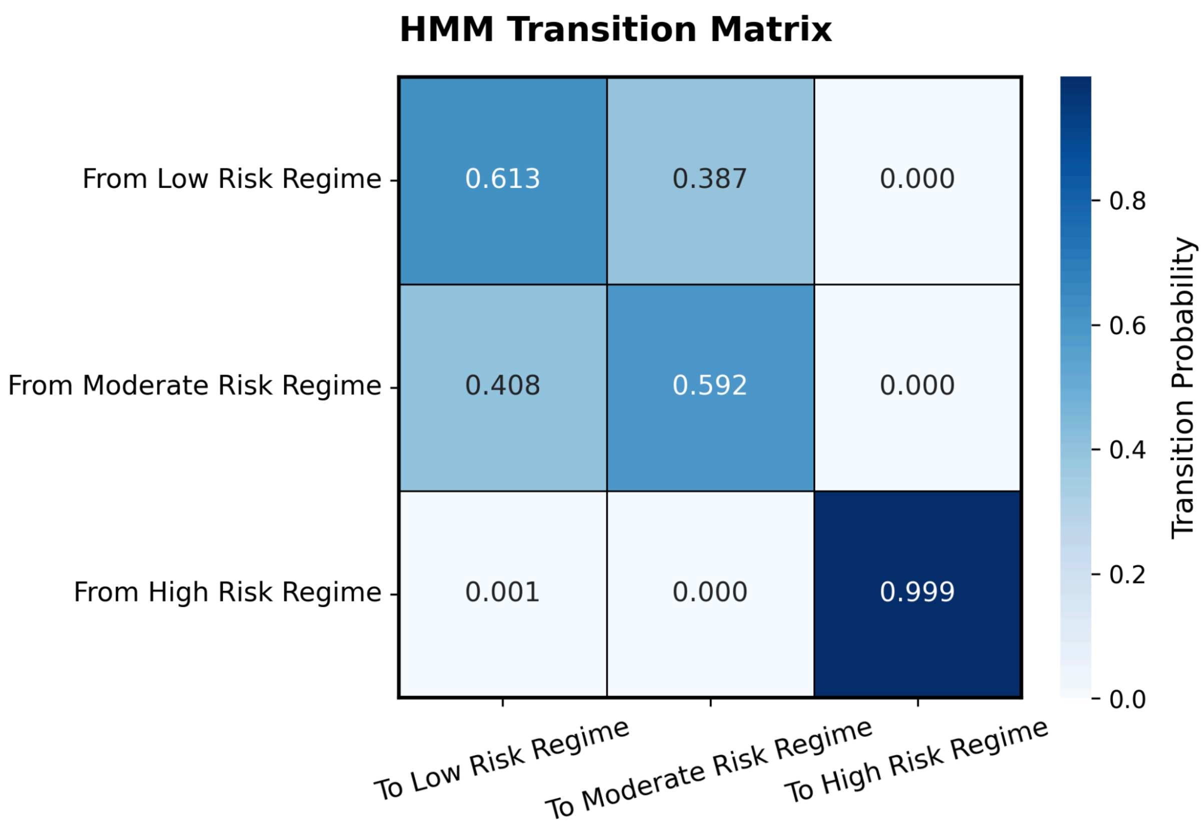
Figure 3 presents the transition matrix, which shows how frequently the city moves between regimes and how persistent each state tends to be. The high-Risk regime is the most persistent state, remaining in place on nearly all consecutive days once it was entered. Low- and Moderate-Risk regimes are less stable and transition into each other more often, reflecting how weather events and short-term fluctuations in crash activity can shift citywide conditions. These weather-driven shifts are consistent with broader findings that precipitation, visibility, and traffic volume influence crash patterns at the city scale (Perrels et al., 2015).

Integrated Insights

Taken together, the three modeling components provide a layered view of crash risk that supports SafePath’s operational goals. The LightGBM model identifies which segments are most likely to experience elevated conditions on a given day, allowing planners to focus attention on a small subset of the network. The GMM clusters explain why certain segments repeatedly appear as high-risk by highlighting the geometric and structural characteristics associated with persistent crash patterns. The HMM regimes add a citywide context layer, indicating whether elevated segment-level scores reflect isolated hotspots or broader system-wide conditions.

This integrated structure helps users interpret daily predictions in context. On High-Risk regime days, elevated scores may reflect conditions affecting the entire network. On Low-Risk days, even moderate scores may be more meaningful because fewer segments exhibit elevated conditions overall. By combining segment-level predictions, long-term structural archetypes, and temporal regimes, SafePath connects short-term risk with broader spatial and environmental patterns, supporting more informed and targeted decision-making.

Figure 3. HMM Regime Transition Probabilities



Dashboard

The SafePath Dashboard provides an integrated interface for exploring crash risk across the city and translating the modeling outputs into an operational tool for daily decision-making. The dashboard is organized into two main tabs, Crash Map and Insights, which together allow users to examine where risk is concentrated, why certain segments exhibit elevated conditions, and when citywide patterns shift.

The Crash Map tab offers an interactive spatial view of both historical crash patterns and model-estimated risk. Users can switch between three map modes: Historical Crash Map, Predicted Risk Map, and Infrastructure Patterns Map. Each mode highlights a different aspect of segment-level conditions. The Historical Crash Map displays observed crash concentrations, while the Predicted Risk Map shades segments by their estimated risk and includes filters for the top 1, 5, and 10 percent of highest-risk segments. The Infrastructure Patterns Map allows users to visualize roadway characteristics such as curvature, visibility-risk, maximum speed, and intersection complexity. Across all modes, selecting a segment opens a details panel that summarizes its crash history, predicted risk, and key infrastructure features.

The Insights tab combines the three modeling components into a structured narrative organized around Why, Where, and When. The Why section presents the feature importance chart and explains the main drivers of predicted risk. The Where section summarizes the spatial archetypes identified by the clustering model and incorporates two additional visual comparisons, the cluster feature heatmap and the risk by speed band chart, to show how different roadway environments exhibit distinct patterns of crash activity. The When section presents the temporal regimes identified by the Hidden Markov Model and shows how citywide conditions shift between baseline, elevated, and weather-driven states. A final panel provides practical takeaways and clarifies the appropriate scope of interpretation.

Together, the Crash Map and Insights tabs allow users to move from high-level patterns to segment-specific details. The dashboard is designed to support diagnosis, prioritization, and situational awareness, while leaving room for professional judgment and field verification. It is deployed using Cloudflare Pages with an R2 bucket, a free-tier, S3-compatible storage service.

Discussion

The modeling results show that crash risk in the city is shaped by a combination of structural, temporal, and environmental factors. The supervised model highlights that risk is not driven by recent crash history alone. Structural features such as curvature, visibility, and turning complexity consistently appear among the strongest predictors, which aligns with longstanding roadway safety principles that geometry and context influence how drivers interact with the network.

The spatial clustering analysis reinforces this point by showing that high-risk segments do not form a single type of roadway environment. Instead, the network contains several recurring segment archetypes, each with its own structural profile and typical crash behavior. These differences help explain why two segments with similar crash histories may behave differently in the future and suggest that interventions should be tailored to the roadway context rather than applied uniformly.

The temporal regime model adds a complementary perspective by showing that citywide conditions shift between baseline, elevated, and weather-driven states. These regimes influence how risk is distributed across the network on any given day, indicating that daily predictions reflect both segment-level characteristics and broader environmental conditions.

Together, these components provide a more complete view of why, where, and when risk emerges. The system is not intended to identify causal mechanisms, but it does surface patterns that can support prioritization and help planners focus attention on segments and periods where risk is likely to be elevated. This layered perspective underscores the value of using SafePath as a diagnostic and exploratory tool that complements, rather than replaces, professional judgment.

Limitations

SafePath provides a structured way to explore patterns in crash risk, but several limitations should be acknowledged. The models identify associations rather than causal effects, and their outputs should not be interpreted as estimates of how specific interventions would change crash outcomes. The analysis relies on reported crash data, which may contain spatial inaccuracies, underreporting, or inconsistencies across time and location. The temporal regime model operates at the daily level and does not capture finer-grained dynamics such as rush-hour patterns or short-duration weather events. Finally, the models do not incorporate temporary changes in traffic patterns, construction activity, or local context that may influence risk on specific days. These limitations reinforce that SafePath should be used as a prioritization and diagnostic tool rather than a prescriptive system.

Ethical Considerations

A key ethical concern for a system like SafePath is the risk of overreliance on model predictions. The system highlights patterns in the data, but it does not establish causality or capture all factors that influence crash risk. If the outputs were treated as definitive or used without additional validation, they could lead to decisions that overlook important local context or temporary conditions not reflected in the data.

SafePath is intended as an exploratory tool that supports interpretation and helps identify locations that may warrant further investigation. Any use of these results for resource allocation or policy decisions should be paired with field verification and professional judgment.

Statement of Work

Both team members contributed to the design, development, and interpretation of the SafePath system. Tanzim Chowdhury led the data pipeline, feature integration, model development, and dashboard implementation. Kevin Leander led the weekly deliverables, final report writing, visual creation, and interpretation of model outputs for the written findings. Both team members contributed to data acquisition, refinement of insights, and overall project direction with guidance from the project team mentor.

Project Resources

GitHub Repository: Team-SafePath/SafePath: SIADS 699 Capstone Project
Jira: Summary - SafePath Project - Jira

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Appendix A: Model HyperParameters

Table A1. LightGBM Parameters

Hyperparameter	Value
Objective	binary
Boosting_type	gbdt
N_estimators	500
Learning_rate	0.05
Num_leaves	31
Max_depth	-1
Min_child_samples	50
Subsample	0.8
Colsample_bytree	0.8
Reg_alpha	0.0
Reg_lambda	0.0
Scale_pos_weight	Derived*
Random_state	42
N_jobs	-1
Verbosity	-1

*Computed as the ratio of negative to positive examples in the training set (non-crash segment-days divided by crash segment-days). The value was calculated dynamically and can be easily reproduced in the modeling scripts.

Table A2. GMM Parameters

Hyperparameter	Value
N_components	5
Covariance_type	full
Random_state	42
N_init	10

Table A3. HMM Parameters

Hyperparameter	Value
N_components	35
Covariance_type	full
N_iter	300
Random_state	42

Appendix B: Additional LightGBM Model Evaluation

Appendix B contains supporting evaluation materials for the LightGBM model. Table B1 reports ROC-AUC, balanced accuracy, average precision, and F1 scores for the validation and test sets. Table B2 presents the raw confusion matrix counts, and Figure B1 shows the normalized confusion matrix derived from the same predictions. These diagnostics complement the Recall@K evaluation by summarizing the model's overall classification performance.

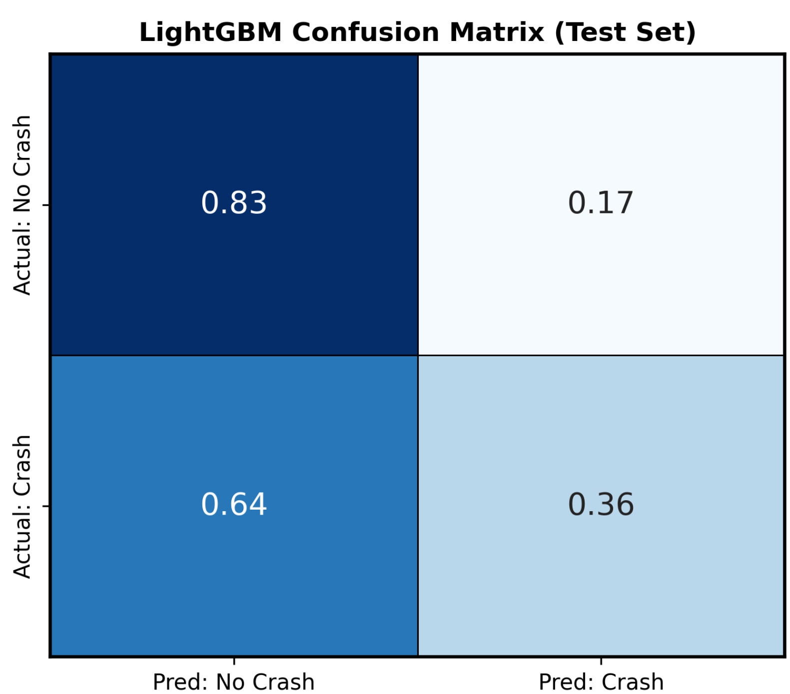
Table B1. LightGBM Performance Metrics Across Validation and Test Sets

Set	**ROC-AUC**	**Average Precision**	**Balanced Accuracy**	**F1 Score**
Validation	0.62162	0.02622	0.59028	0.04574
Test	0.62982	0.02517	0.59484	0.04322

Table B2. Raw Confusion Matrix Counts (LightGBM Test Set)

	Predicted: No Crash	**Predicted: Crash**
Actual: No Crash	3517056	697311
Actual: Crash	29818	16241

Figure B1. Normalized Confusion Matrix for the LightGBM Model



Appendix C: Cluster and Regime Summaries

Table C1 reports the key characteristics of the five segment-level clusters, including crash metrics, geometric features, and exposure-related attributes. Figure C1 provides a normalized feature heatmap that highlights how these clusters differ across roadway geometry, visibility, speed environment, and intersection structure. The dashboard version additionally includes predicted-risk for stakeholders who want to explore how model outputs vary across cluster types.

Table C1: Summary of GMM Segment Clusters

Feature	**Cluster 1**	**Cluster 2**	**Cluster 3**	**Cluster 4**	**Cluster 5**
N_Segments	1526	340	2606	5343	1826
Total Crashes	44.705111	121.626471	32.922487	44.676961	63.012596
Avg Crash Rate	0.017062	0.043176	0.012680	0.017079	0.023516
Crash Volatility	0.124148	0.178647	0.110159	0.124274	0.139281
% of Days with a Crash	0.017062	0.043176	0.012680	0.017079	0.023516
Curvature	0.018813	0.282470	0.000041	0.000036	0.009124
Visibility	0.150028	0.163078	0.114534	0.115986	0.093679
Intersections	4.354522	3.720588	3.945894	4.000000	3.000000

Figure C1: Segment Cluster Feature Heatmap

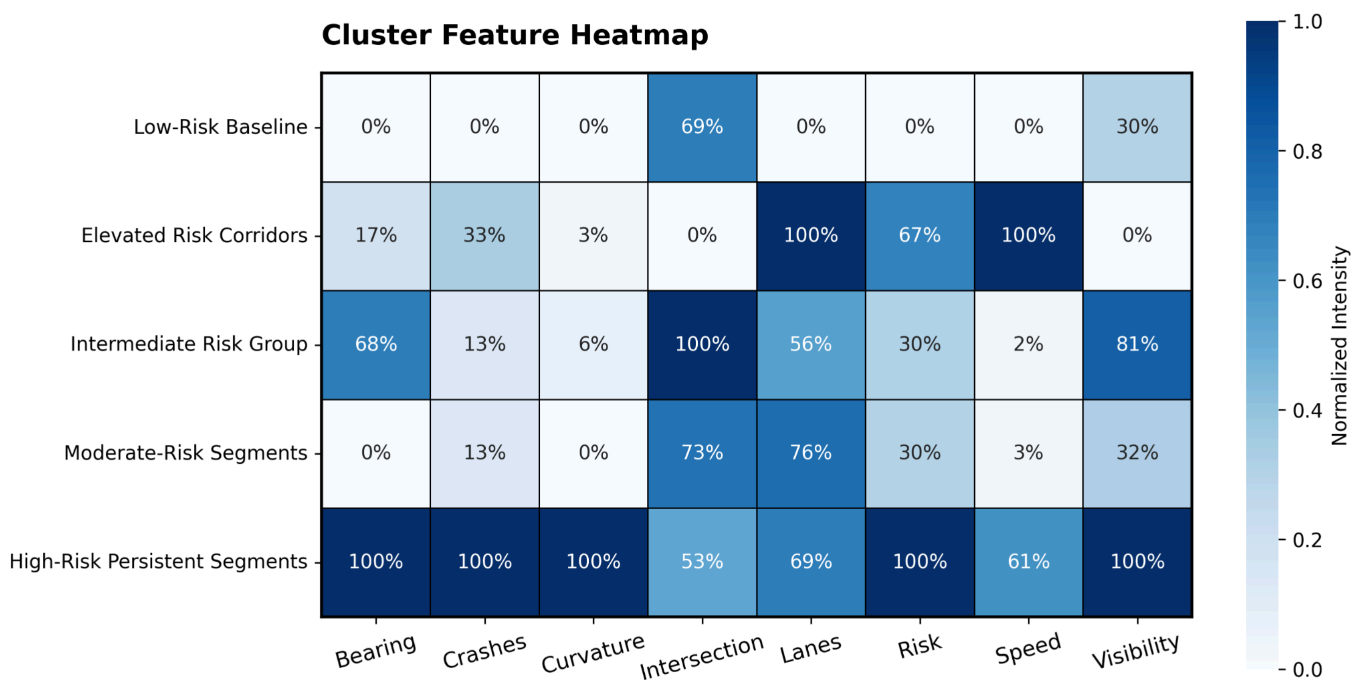


Table C2 summarizes the three daily risk regimes identified by the Hidden Markov Model. These regimes reflect broader citywide conditions shaped by weather patterns and recent crash activity, and they provide the temporal context used to interpret day-to-day variation in segment-level risk.

Table C2: Summary of HMM Daily Risk Regimes

Feature	**Low Risk Regime**	**Moderate Risk Regime**	**High Risk Regime**
Number of Days	885	839	833
Total Crashes	149.22410	150.10583	350.71135
Crash Rate	0.01257	0.01265	0.02884
Average Crashes last 7 days	0.09045	0.08945	0.21068
Average Crashes last 30 days	0.38875	0.38294	0.89853
Temperature	14.51453	12.01605	11.25012
Precipitation	7.41047	0.00000	4.06534
Windspeed	22.57128	20.26266	21.89275
Rain Indicator	1.00000	0.00000	0.58696

Appendix D: Variable Dictionary

Table D1. Full List of Variables with Descriptions

Variable	Type	Description
Identifiers		
segment_id	String	Unique identifier for each street segment (derived from OSM [u, v, key] variables
date	Date	Calendar date for the observation, used at both the segment-day level and city-day level
Crash Metrics		
crash_count	Integer	Number of crashes that occurred on this segment on this date
crash_occurred	Binary (0/1)	Indicator for whether at least one crash occurred on this segment on this date
total_crashes	Integer	Total number of crashes, defined per segment across the study period in segment-level summaries and per city-day in the HMM dataset
avg_crash_rate	Float	Average crash rate, computed per segment across the study period in segment-level summaries
crash_volatility	Float	Variability in daily crash counts for a segment over time
pct_days_with_crash	Float (0-1)	Proportion of days in the study period on which the segment experienced at least one crash
crash_rate	Float	Daily crash rate aggregated at the city level in the HMM dataset
crash_rate_rain	Float	Crash rate for a segment on rainy days, computed in segment-level summaries
crash_rate_no_rain	Float	Crash rate for a segment on non-rainy days, computed in segment-level summaries
Weather Features		
temperature_2m_mean	Float	Mean daily temperature, recorded at the segment-day level in the main panel and aggregated at the city level in the HMM dataset
avg_temp	Float	Average daily temperature across all days for a segment, used in segment-level summaries
percipitation_sum	Float	Total daily precipitation, recorded at the segment-day level in the main panel and aggregated at the city level in the HMM dataset

Variable	Type	Description
avg_percip	Float	Average daily precipitation across all days for a segment, used in segment-level summaries
windspeed_10m_max	Float	Maximum daily windspeed, recorded at the segment-day level in the main panel and aggregated at the city level in the HMM dataset
avg_wind	Float	Average daily maximum windspeed across all days for a segment, used in segment-level summaries
rain_indicator	Binary (0/1)	Indicator for whether measurable precipitation occurred, defined at both the segment-day level and city-day level
Segment Geometry		
segment_length	Float	Length of the street segment in meters
Calendar Features		
day_of_week	Integer (0-6)	Day of week for the observation(0 = Monday)
month	Integer (1-12)	Month of year for the observation
sin_month	Float	Sine-encoded month for seasonal patterns
cos_month	Float	Cosine-encoded month for seasonal patterns
sin_day_of_week	Float	Sine-encoded day of week for seasonal patterns
cos_day_of_week	Float	Cosine-encoded day of week for seasonal patterns
Lag Features (Crash History)		
crashes_last_7_days	Integer	Number of crashes on this segment in the previous seven days
avg_crashes_last_7_days	Float	Average number of crashes in the previous seven days, computed at the segment level in segment summaries and at the city level in the HMM dataset
crashes_last_30_days	Integer	Number of crashes on this segment in the previous thirty days
avg_crashes_last_30_days	Float	Average number of crashes in the previous thirty days, computed at the segment level in segment summaries and at the city level in the HMM dataset
Model Outputs		
predicted_proba	Float (0-1)	Predicted probability of at least one crash occurring on this segment-day

Variable	Type	Description
Cluster Labels (Segment-Level Clustering)		
kmeans_cluster	Integer	Cluster assignment for each segment from the K-Means clustering model
gmm_clustering	Integer	Cluster assignment for each segment from the Gaussian Mixture Model
HMM-Specific Variables		
hidden_state	Integer	Latent state index inferred by the Hidden Markov Model, representing underlying crash risk regimes
state_probability_max	Float (0-1)	Maximum posterior probability of the most likely hidden state for that date
state_label	Categorical	Human-readable label for the hidden state (e.g., “Low risk”, “Medium risk”, “High risk”)
Road Type Features		
These variables represent one-hot encoded OpenStreetMap (OSM) road classifications. Each variable equals 1 if the segment’s OSM tags include the corresponding road type or combination of types. All variables in this section share the same type and description.		
road_busway	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_living_street	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_living_street residential	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_motorway	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_motorway motorway_link	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_motorway primary	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_motorway tertiary motorway_link	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_motorway_link	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_motorway_link motorway	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_motorway_link primary	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_motorway_link residential	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)

Variable	Type	Description
road_motorway_link secondary	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_motorway_link tertiary	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_motorway_link unclassified	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_primary	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_primary_link	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_primary_link primary	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_primary_link trunk_link	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_residential	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_secondary	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_secondary living_street	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_secondary primary	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_secondary residential	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_secondary secondary_link	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_secondary tertiary	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_secondary_link	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_secondary_link residential	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_secondary_link unclassified	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_tertiary	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_tertiary residential	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_tertiary_link	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)

Variable	Type	Description
road_tertiary_link residential	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_tertiary_link tertiary	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_tertiary_link unclassified	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_trunk	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_trunk_link	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_unclassified	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_unclassified living_street	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_unclassified residential	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)
road_unclassified tertiary	Binary (0/1)	Binary indicator derived from OpenStreetMap (OSM)